

PROJECT REPORT



**College Finance & Expense Management System
(CFEMS)**

CSE2006-PROGRAMMING IN JAVA

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Introduction

Colleges handle various financial activities such as income, expenses, budgets, vendor payments, and donations. Managing these records manually can make tracking, searching, and reporting difficult.

The College Finance & Expense Management System (CFEMS) is a Java-based desktop application that provides a centralized system for managing transactions, budgets, vendors, payments, and financial reports.

The application uses Java Swing for the GUI, MySQL for data storage, and JDBC for database connectivity. It also demonstrates Java concepts such as OOP, inheritance, polymorphism, abstraction, exception handling, enums, multithreading, and synchronization.

Problem Statement

Managing college finances manually can make it difficult to track **income, expenses, budgets, vendors, and payments**. It may also lead to overspending, difficulty in finding transactions, and errors in financial calculations.

The **College Finance & Expense Management System (CFEMS)** provides a centralized solution to record, manage, search, and report financial data. It also validates expenses against budgets, manages payment statuses, and stores records in a **MySQL database**, reducing manual effort and keeping financial information organized.

Functional Requirements

User Authentication

The system provides role-based login for **Admin, Accountant, and Viewer**, with different access permissions.

Dashboard

Displays important financial information such as **total income, expenses, balance, and transactions**.

Income Management

Allows users to record income with **amount, category, description, date, and source**.

Expense Management

Allows users to add and categorize expenses with relevant transaction details.

Budget Management

Allows creation and tracking of budgets and prevents expenses from exceeding the available budget.

Vendor Management

Stores vendor details such as **ID, name, contact, and category**.

Payment Management

Manages payments using statuses such as **Pending, Partial, Paid, and Cancelled**, while preventing overpayment.

Transaction Management

Stores and manages income and expense transactions with their relevant details.

Transaction Search

Allows users to search transactions using details such as **ID, type, category, or vendor**.

Financial Reports

Provides summaries of **income, expenses, balance, transaction count, and category-wise spending**.

File Export

Allows transaction records to be exported to a **transactions.txt** file.

Non-Functional Requirements

Usability

The system has a simple and organized interface that is easy to use.

Reliability

The system validates financial operations and prevents invalid actions such as budget overflow and overpayment.

Data Persistence

Important records are stored permanently in the **MySQL database**.

Maintainability

The application is divided into separate **GUI, logic, DAO, and model classes**, making it easier to modify and maintain.

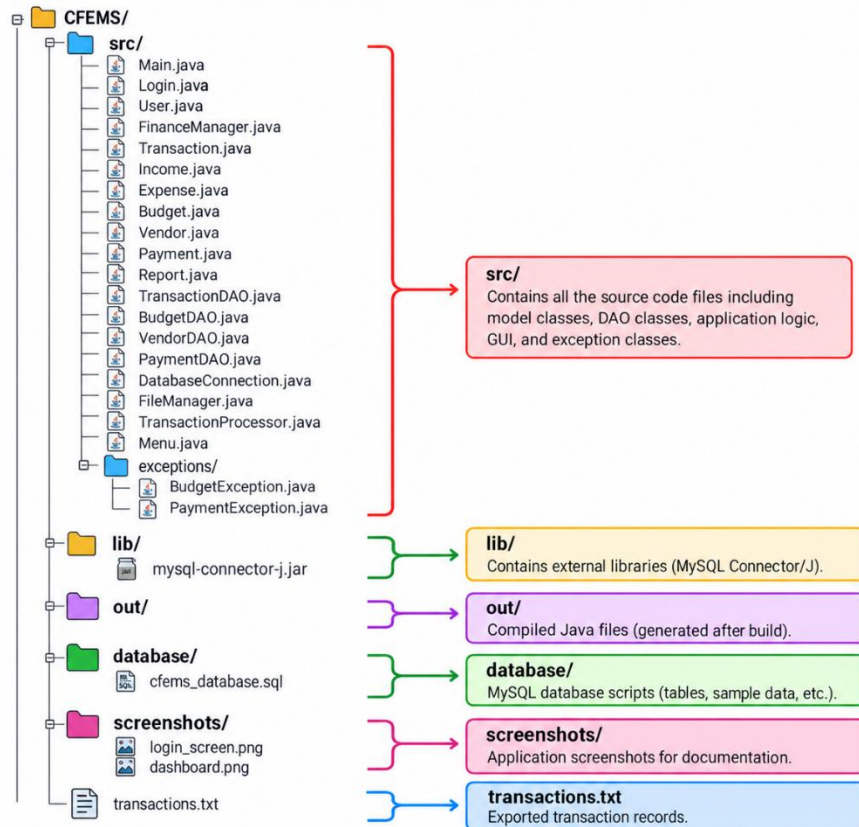
Performance

The system provides quick responses for common operations and uses prepared statements for database access.

Security

User authentication is implemented through role-based login, and database credentials should be kept private.

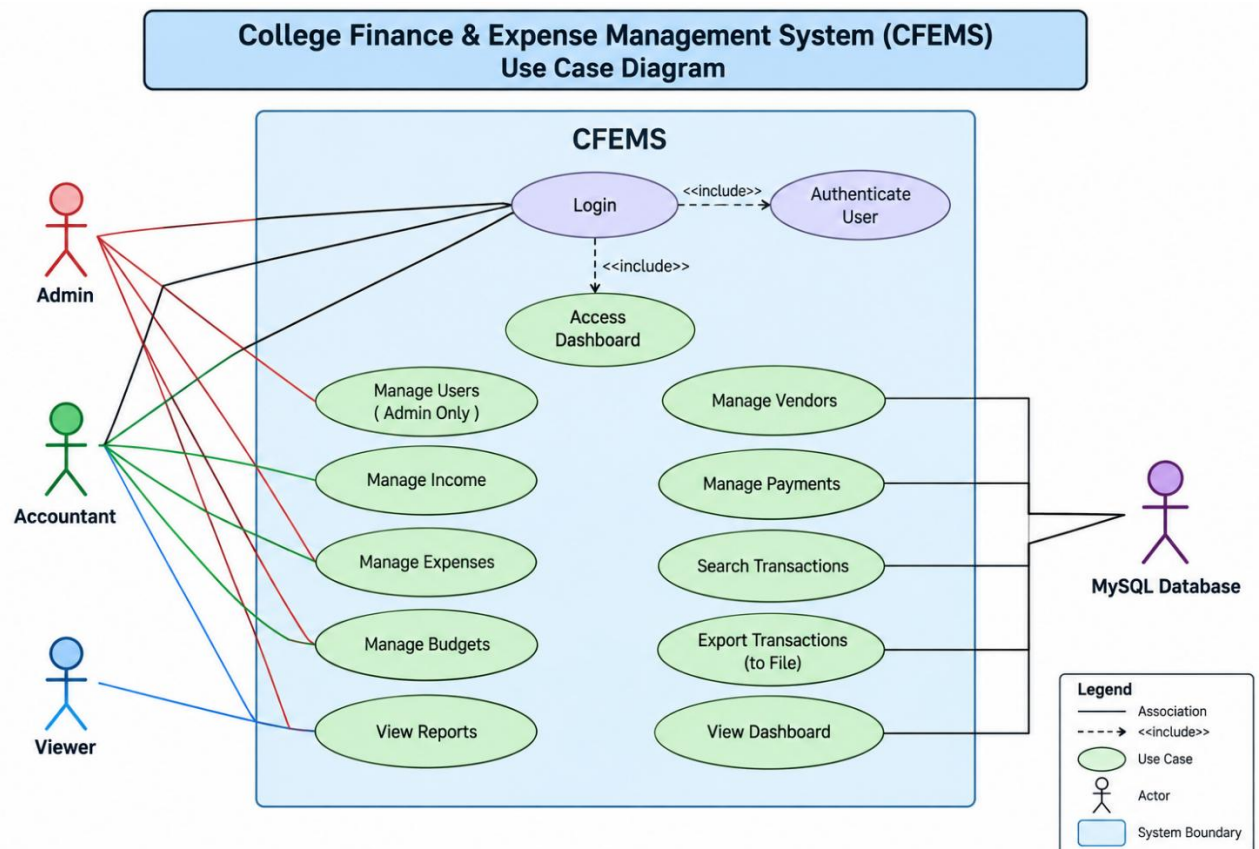
Project Structure



Design Diagrams

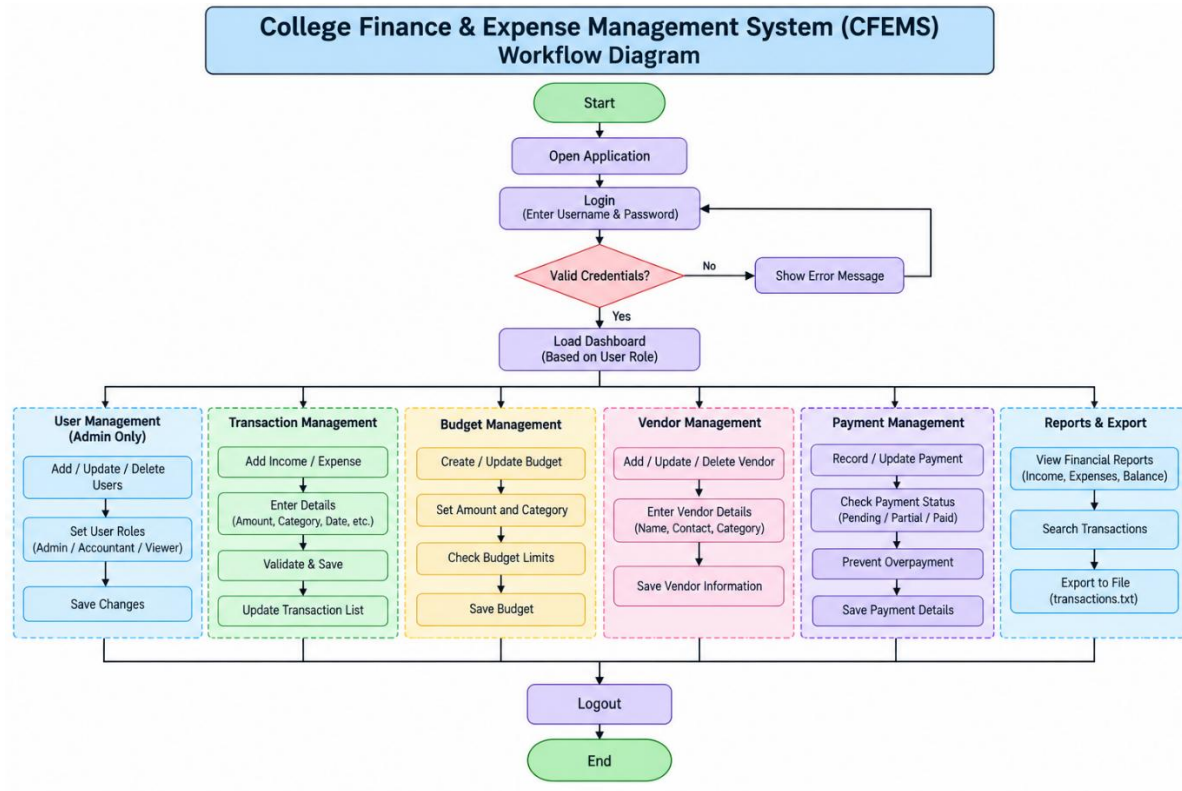
1-Use Case Diagram

The Use Case Diagram shows how different users such as Admin, Accountant, and Viewer interact with CFEMS. It represents major functions like login, transactions, budgets, payments, vendors, and reports.



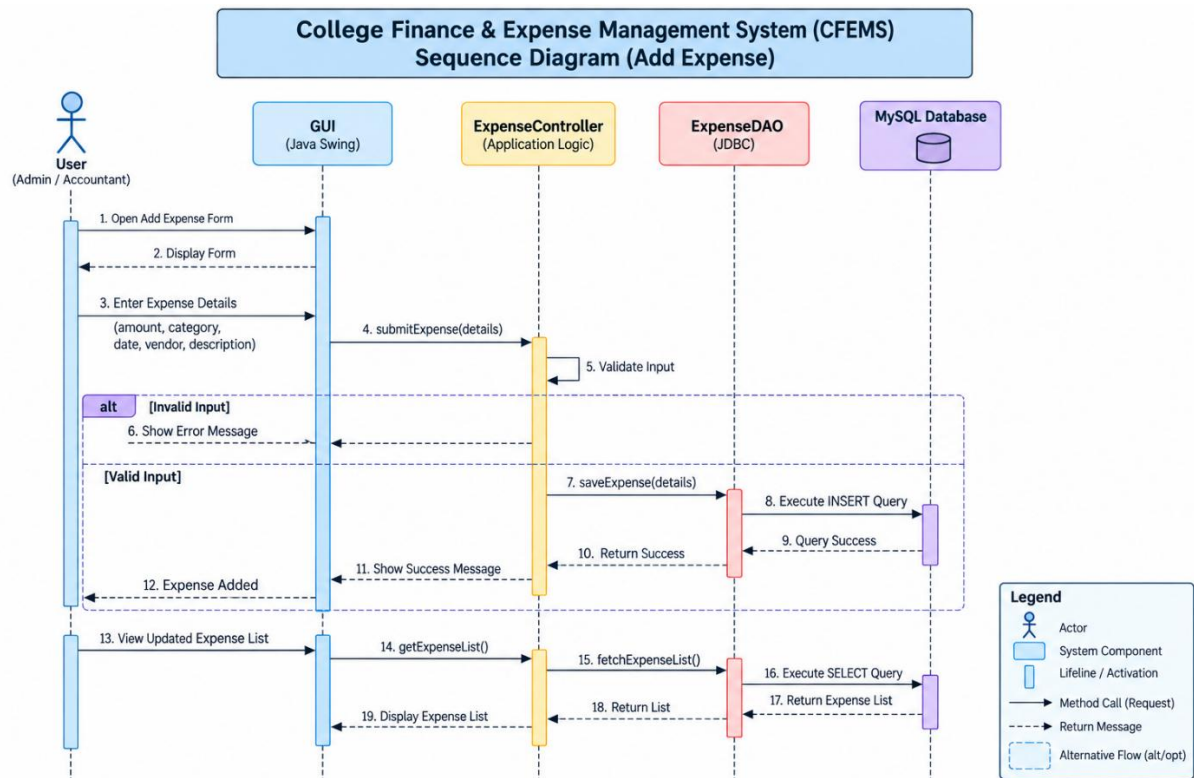
2-Workflow Diagram

The Workflow Diagram shows the step-by-step flow of the system, starting from login and continuing through different financial operations until logout.



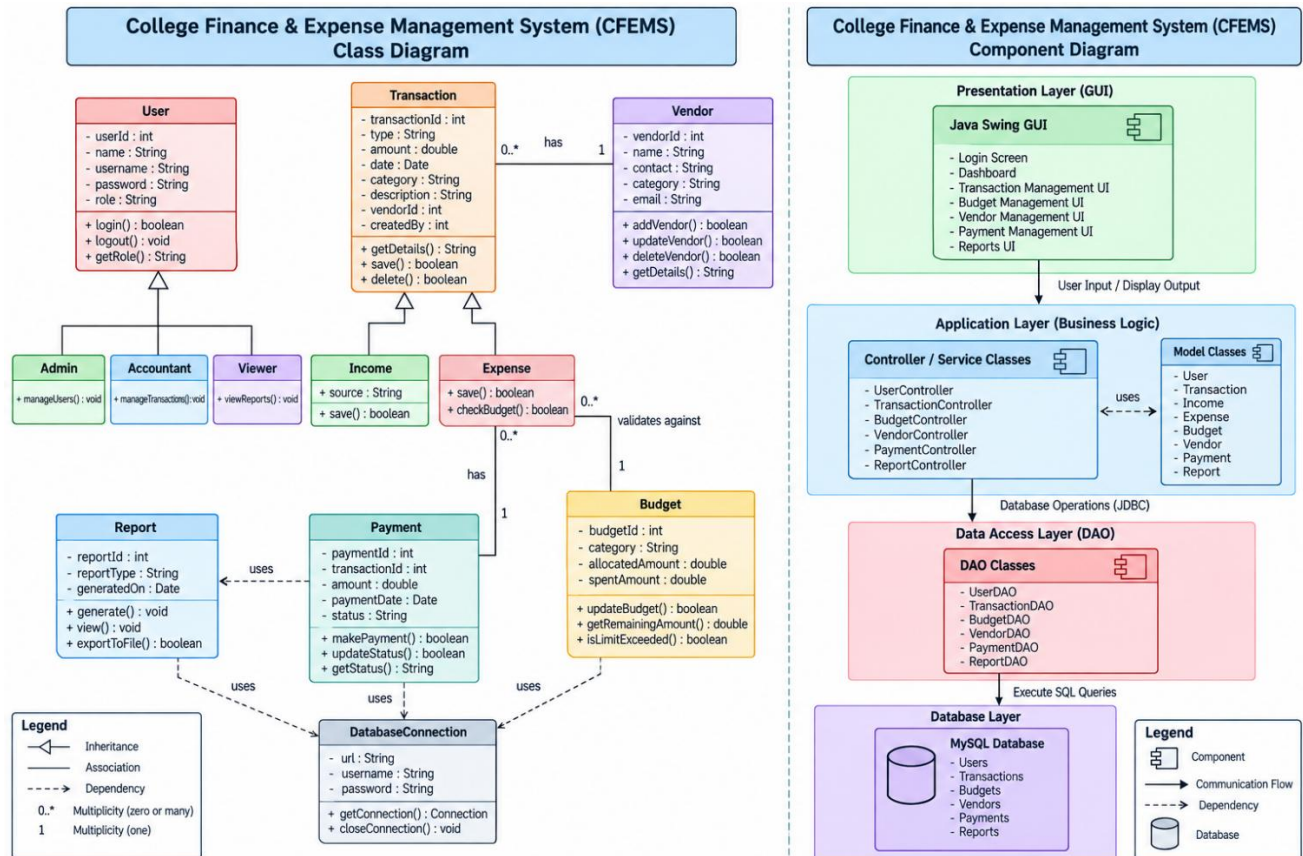
3-Sequence Diagram

The Sequence Diagram shows how different components of CFEMS communicate with each other over time while performing an operation such as adding an expense.



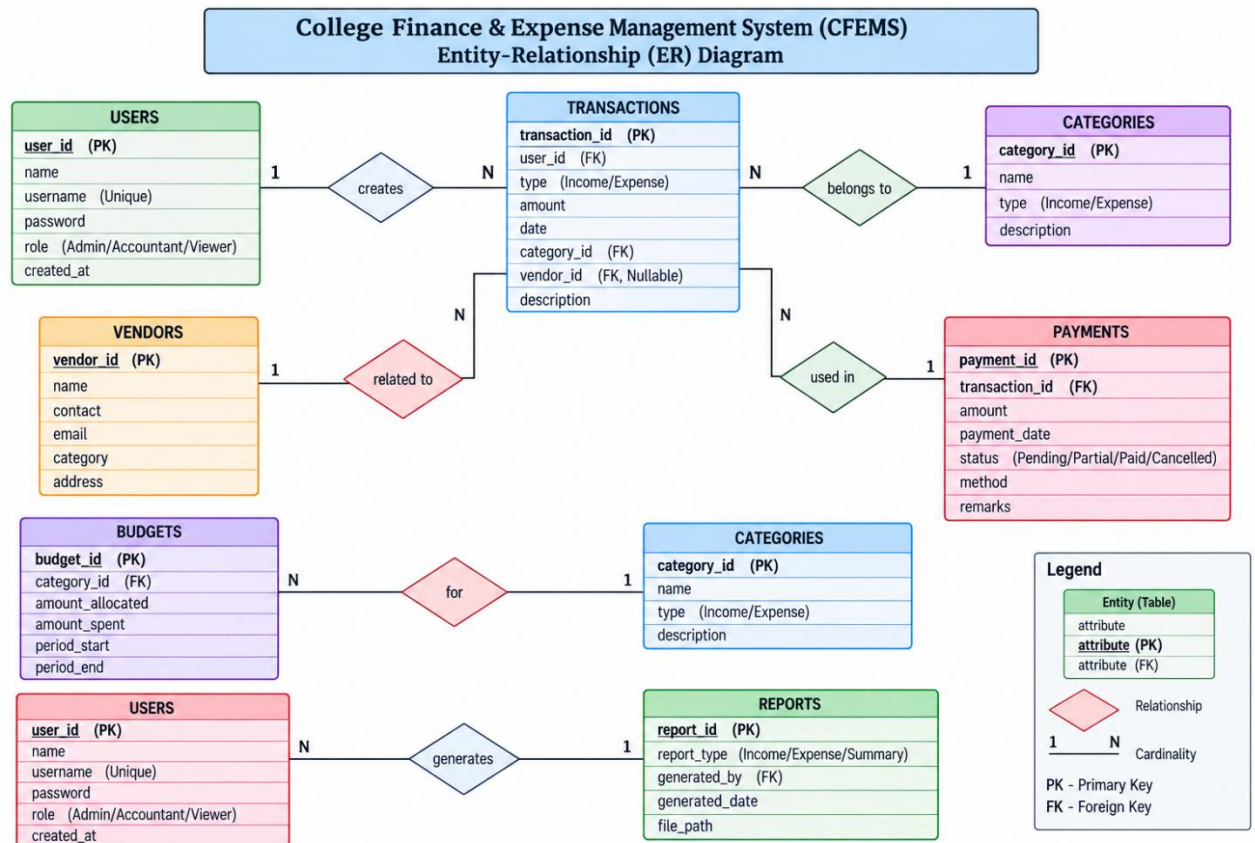
4-Class/Component Diagram

The Class Diagram represents the main Java classes, their attributes, methods, and relationships. The Component Diagram shows how the GUI, application logic, DAO layer, and database work together.



5 ER Diagram

The ER Diagram represents the database structure of CFEMS, including tables such as users, transactions, budgets, vendors, and payments, along with their relationships.



Design Decisions & Rationale

Java Swing

Java Swing was used to create a simple desktop GUI for easy interaction with the system.

MySQL Database

MySQL was selected to store financial records permanently and keep the data organized.

JDBC

JDBC is used to connect Java with MySQL and perform database operations through SQL queries.

DAO Pattern

Separate DAO classes were used to keep database operations separate from the main application logic.

Role-Based Access

Different roles such as Admin, Accountant, and Viewer were implemented to control access to system features.

Exception Handling

Custom exceptions and validation are used to handle invalid amounts, budget limits, and payment errors safely.

Multithreading & Synchronization

Multithreading is included to demonstrate concurrent transaction processing, while synchronization helps prevent conflicts during shared operations.

Implementation Details

Programming Language

The system is implemented using **Java** and follows object-oriented programming principles.

GUI

Java Swing is used to create the login screen, dashboard, and different management modules.

Database

MySQL is used to store users, transactions, budgets, vendors, and payment records.

Database Connectivity

JDBC is used to establish the connection between Java and MySQL and perform CRUD operations.

OOP Concepts

The project implements **encapsulation, inheritance, polymorphism, abstraction, interfaces, and enums** through different Java classes.

Exception Handling

Custom exceptions and validation are used to handle invalid amounts, budget limits, and payment-related errors.

Multithreading

Runnable and **Thread** are used for transaction processing, with synchronization to safely handle shared operations.

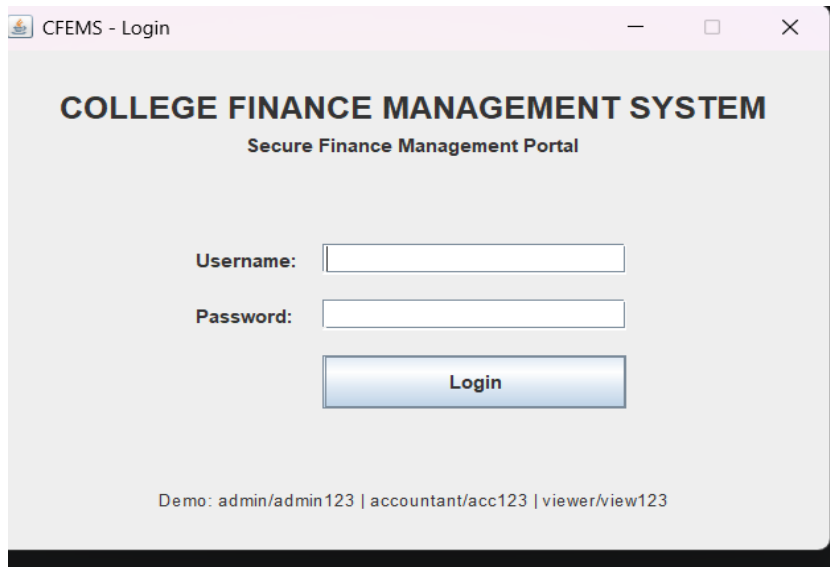
File Export

Transaction records can be exported to a **transactions.txt** file for simple external access and record keeping.

Screenshots / Results

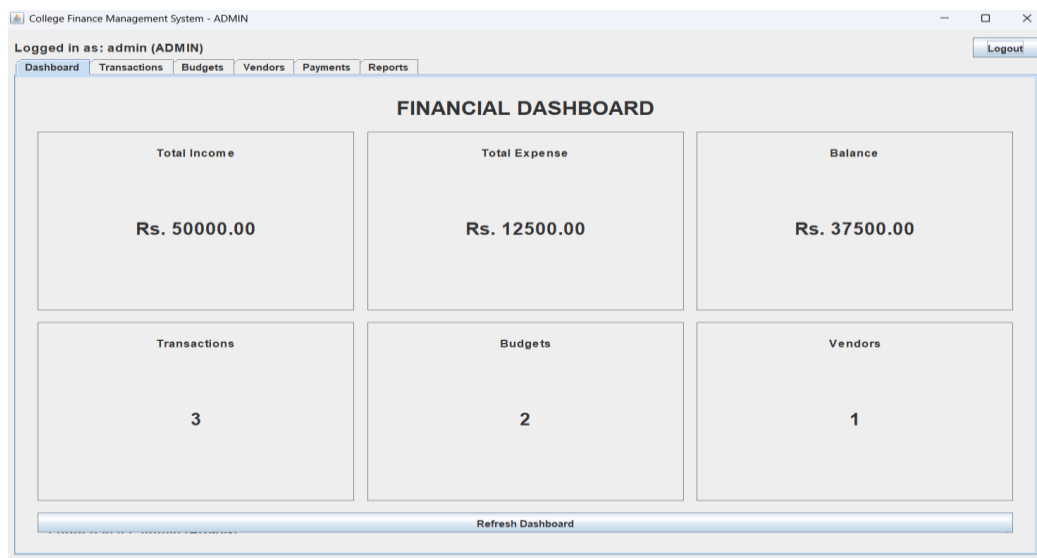
This section presents screenshots of the working CFEMS application.

9.1 Login Screen



The login screen asks the user to enter a username and password. Based on the credentials entered, the application identifies the user's role (Admin, Accountant, or Viewer) and opens the dashboard accordingly.

9.2 Dashboard



The dashboard gives a quick overview of the college's finances after login, showing the total income, total expense, balance, and the number of transactions, budgets, and vendors currently recorded in the system.

9.3 Budget Management

The screenshot shows the 'Budgets' tab in the 'College Finance Management System - ADMIN'. The user is logged in as 'admin (ADMIN)'. The interface includes a 'Logout' button and a navigation bar with 'Dashboard', 'Transactions', 'Budgets', 'Vendors', 'Payments', and 'Reports'. The 'Budgets' section has a form with 'Category' and 'Budget Amount' input fields, a 'Create Budget' button, and a 'Clear' button. Below the form is a table with columns: ID, Category, Budget, Spent, Remaining, and Utilization. The table contains two rows of data.

ID	Category	Budget	Spent	Remaining	Utilization
1	transport	15000.00	0.00	15000.00	0.00%
2	food	50000.00	10000.00	40000.00	20.00%

This screen shows the budget module, where a category-wise budget has been created. It displays the budget amount, the amount spent, the remaining amount, and the utilization percentage for each category.

9.4 Vendor Management

The screenshot shows the 'Vendors' tab in the 'College Finance Management System - ADMIN'. The user is logged in as 'admin (ADMIN)'. The interface includes a 'Logout' button and a navigation bar with 'Dashboard', 'Transactions', 'Budgets', 'Vendors', 'Payments', and 'Reports'. The 'Vendors' section has a form with 'Vendor Name', 'Category', and 'Contact' input fields, an 'Add Vendor' button, and a 'Clear' button. Below the form is a table with columns: ID, Name, Contact, and Category. The table contains one row of data.

ID	Name	Contact	Category
1	abc	9839658970	electricity

This screen shows the vendor module, where a new vendor can be added with a name, contact number, and category. The table below the form lists the vendors already added to the system.

9.5 Payment Management

College Finance Management System - ADMIN

Logged in as: admin (ADMIN) [Logout](#)

Dashboard Transactions Budgets Vendors Payments Reports

Vendor Total Amount [Add Payment](#)

Payment ID Amount [Make Payment](#) [Cancel Payment](#)

ID	Vendor	Total	Paid	Status
3	adarsh	10000.00	10000.00	PAID
2	abs	1000.00	0.00	PENDING
1	abs	1000.00	0.00	PENDING

This screen shows the payment records along with their payment status. The table lists each payment's vendor, total amount, amount paid so far, and its current status, such as PAID or PENDING.

9.6 Financial Reports

College Finance Management System - ADMIN

Logged in as: admin (ADMIN) [Logout](#)

Dashboard Transactions Budgets Vendors Payments Reports

Total Income Rs. 50000.00 Total Expense Rs. 12500.00 Balance Rs. 37500.00 Transactions 3

Expense Category	Total Expense
food	10000.00
electricity	2500.00

[Refresh Report](#)

This screen shows the financial report generated by the application. It displays the total income, total expenses, balance, and the number of transactions, along with a category-wise breakdown of expenses.

Testing Approach

The CFEMS application was tested by performing different operations and checking whether the system produced the expected results.

Login Testing

Different user credentials were tested to verify role-based access for Admin, Accountant, and Viewer.

Functional Testing

Income, expenses, budgets, vendors, payments, search, and reports were tested individually.

Validation Testing

Invalid inputs such as negative amounts, exceeding budgets, and overpayments were tested to ensure they were rejected.

Database Testing

Records were added, updated, and retrieved to verify proper communication between Java and the MySQL database.

File Export Testing

Transaction data was exported to transactions.txt and checked for correct output.

Challenges Faced

Database Connectivity

Connecting the Java application with MySQL using JDBC and handling database errors was initially challenging.

GUI Development

Designing the Swing interface and connecting different buttons and modules with the required functions required careful implementation.

Data Validation

Implementing checks for invalid amounts, budget limits, and payment overlimits required additional validation and exception handling.

Role-Based Access

Managing different permissions for Admin, Accountant, and Viewer required proper access control in the application.

Multithreading

Implementing threads and synchronization for transaction processing required understanding how shared data should be handled safely.

Conclusion

The **College Finance & Expense Management System (CFEMS)** provides a simple and centralized way to manage college income, expenses, budgets, vendors, and payments. The project successfully demonstrates practical use of **Java, OOP, Swing, JDBC, MySQL, exception handling, and multithreading** in a real-world application.